

Short Description

My little garden project is a react-native app that runs locally and can run offline (with exception of the AI suggestion). It is built to help with gardening and layout places where vegetables and herbs will be planted and grown in the most optimal way with spacing, with companionships between plants and with the microclimate that is created with it.

It takes a lot of influence from the Planter app in the app store, but with the addition of being more customizable and having AI do suggestions.

Splash Screen

This is a splash screen that displays a logo (assets/splash.png) and a suitable background.

Environment Screen

On your first time, you will be asked where the sun will come up by choosing one of the 16 directions (a sun icon in the middle and 16 selectable circles/buttons). Once you select one, you get a confirmation button that once pressed, this information is stored and can be configured later in the garden settings screen.

You will also be asked when your frost dates are (spring and fall). This will later determine the growing calender of the plant. (See growing calender section). This can also be changed later in the garden settings screen.

Home Screen

On the home screen, you will see a garden (a green garden like background). In the top right corner you have a settings button that will display what the total size of your garden is (in meters). This will be used to define the size of the components of the Garden relative tot he screen. The screen is, however, moveable so that, should the garden not fit proportionally into the screen, the user can move around according to the best practices for mobile usage.

In the top left corner, there is a list button (See list button section). In the lower right corner there is a floating add (+) button (See add button section).

If you have added components to your garden, these will be displayed on this screen (see components section). If you long press on them you get a pop menu (see components section)

Add Button

The add button hovers at the right bottom of the Home screen. When the user presses on this button, a modal opens.

In this model the user can choose between two components that can be added onto the screen:

- A garden box (for vegetables and herbs). Rectangular.
- A pot. Circle.
- Rectangular tower (consists of multiple layers)
- A Circular tower (consists of multiple layers)

When choosing one you will move on within the modal to define the length/width of the garden box and the radius of the circle. You also define how wide the border is. For the towers, you define the amount of layers and for each layer what the size is together with the border size of each layer. You give the component an unique name (enforced by app). You can always go back tot the previous section.

Once selected and the size filled in, you can placet his garden box into the garden. This is a drag and drop feature and if there are already existing ones, they can be moved as well when moving over them.

You can rotate these by 15°. Once placed you can click on the green checkmark button. You can also cancel (and revert everything you did) by clicking on the red cross button.

There will be validation on the sizes when the available space is not enough in one or two directions with the existing layout you have for the garden.

List Button

The list button will display an overlay that displays in an accordion list all the components you have placed in this garden. It will display the name and the time of creation (which was saved during the creation and placement of the component). When clicking on one of them you get an overview of all the planted herbs and vegetables. You can close this overlay with a x button.

Components

Your garden consists of different components. These are created with the add button at the home screen. They have a name, a creation timestamp, a size, coordinates and a list of plants that can be found in the component.

A garden box and a pot are very basic with just their sizes and border sizes. While both the rectangular tower and the circular tower have different levels with each level their size and border width. You start with the lowest level, which has the biggest size. Then you work up with smaller sizes for each level (the app enforces this).

You can edit components only when they are empty by long pressing on the box/circle in the home screen displayed in your garden. This is only possible if they are empty (enforced by app). You can also delete this way (a delete button is displayed on the screen).

You can then go to a “zoomed-in” version of the components, which is actually a different screen. This is done by clicking on the component in the home screen, or clicking on the “look” icon/button in the accordion list. We call this the “Component Detail Screen”.

You can add plants, by clicking on the add button, similar to the Home Screen. This is the add plant screen (see add plant screen section).

Component Detail Screen

In the component detail screen you get a zoomed in version of the component. This is a rectangular/circle that contains multiple icons, which represents the plants planted. Below this icon there is a circle which represents the minimal space the plant or herb will need. When you long press one of these plants, you can drag/drop the plant similar to how you would drag-n-drop the components in your garden. During the drag-n-drop there will also be a relation showing up between the surrounding plants. Plants can be companion plants, neutral or hostile. This is represented by a green circle (companion) or red circle (hostile) between the plants (in all directions that have a plant. This is also shown when you are not moving any plants. When drag-n-dropping a plant you have an option to cancel or confirm the position. At the bottom of the screen you get a bin icon to remove the plant.

There should also always be a go back button available to go back to the Home Screen.

A List icon, similar to that one of the Home menu, is also available. When clicking on this list icon, you get a list of all the plants you have planted in the garden. Each plant also has a name (does not need to be unique) and a creation date. This is also an accordion and when clicking an action row gets shown below. It has a “zoom” icon that highlights the plant in the component, a move icon, to start moving and drag-n-drop the component and an info icon. The info icon will lead to the plant details screen.

Plants details screen

A plant details screen contains a picture of the plant, their “family/category”, a detailed description. Quick info (see quick info image), a growing calendar, Companion plants (with a relation from giving or receiving a benefit (eg: holding nitrate in the ground) and what they give or receive or both), combative/hostile plants, with a relation of what they do to the other plant (eg: prevents growth) or what the other plant is doing to this plant or both. What pests can happen to this plant, what diseases, and beneficial critters.

There is also an instruction on how to grow it from seed, planting considerations, feeding, harvest, storage and pruning.

Quick info

Quick info is a grid layout with cards with short information on this plant.

Growing Calendar

Growing calendar is based on your frost dates and will present the dates when you need to start inside, transplant and sow outside.

Companion Plants

Companion plants shows a list of other plants which give, receive or do both in relation to this plant.

This is visually good visible.

Combative Plants

The opposition of Companion Plants.

Add Plant Screen

During the add plant screen, you are on a different screen where you can select multiple plants to be added. This is a grid with different tiles and when you click on a tile, you are planning to add this plant (a number 1 appears). You can then add or decrease the amount of plants you can add (to a minimum of 0). If no plants are selected, you go out of the selection mode. A long click or clicking on the info icon on a tile opens the Plant Details Screen.

A tile contains the name of the plant, an icon and the family/category this plant belongs to.

Once you confirm the selection of plants to be added by clicking the confirm button, you can then place one by one a plant. This has already been discussed before how this works.

Non-Technical Requirements

- You use meters and centimeters
- Language is in Dutch. Foresee an i18n file and setup.
- A go back button should always be available
- Destructive actions should always be prompted to be confirmed.
- In the settings there is an export button which saves the complete data (you garden and all the data the app uses) into a json.
- This is user friendly and has a nature-like UI.
- The app should be used on Android

Technical Requirements

- We use React-Native, Nativewind and Zustand as storage. The storage is persisted on the phone (and thus can be used offline). Typescript is our programming language.
- For validation we use ZOD.
- We use KISS, DDD, BDD, non-anemic domain model, a clean architecture, SOLID principles, clean naming which is focused on domain terms and not to technical. Readability is key here.
- The architecture is a clear separation between UI models, domain models and storage/persistent models.
- The UI components are setup in an component folder. They follow the atomic design and are purely dump components. We use BEM notation and place all the nativewind classes in a separate css file, where we use @apply to apply the nativewind utility classes wrapped by a good BEM notated class.
- We use Meastro for testing the interfaces.
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